

DS 4440

Computation Graphs

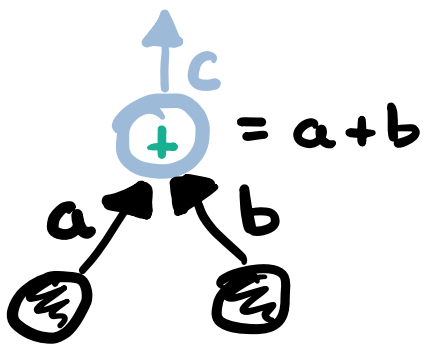
So far:

- Supervised learning
- **SGD** for linear models
- **MLPs** and non-linear models

Today: Computation Graphs as The core abstraction for **Neural Networks**.

The Big Idea ^(TM)

Models as **Directed Acyclic Graphs (DAGs)**



$$c = a + b$$

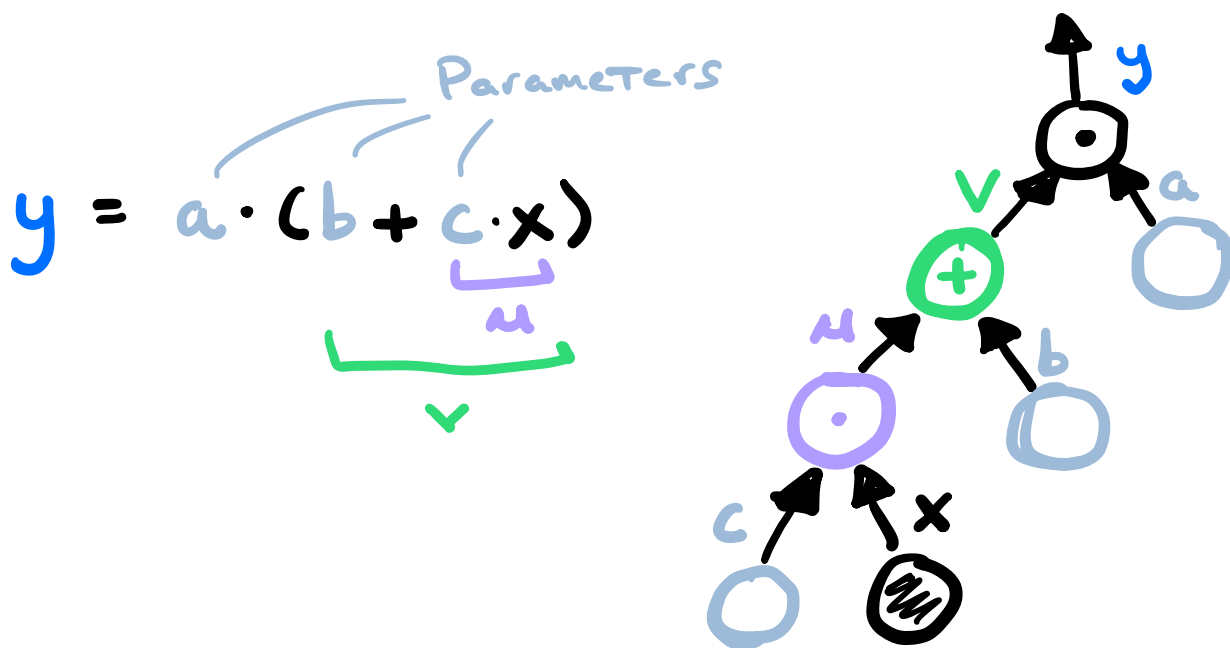
($\approx$) Nodes as **operations** (possibly identity), labeled as **variables**.

Tensors (scalars above) "flow" along

edges —.

We care about **forward** and **backward** Passes Through Such graphs.

Consider



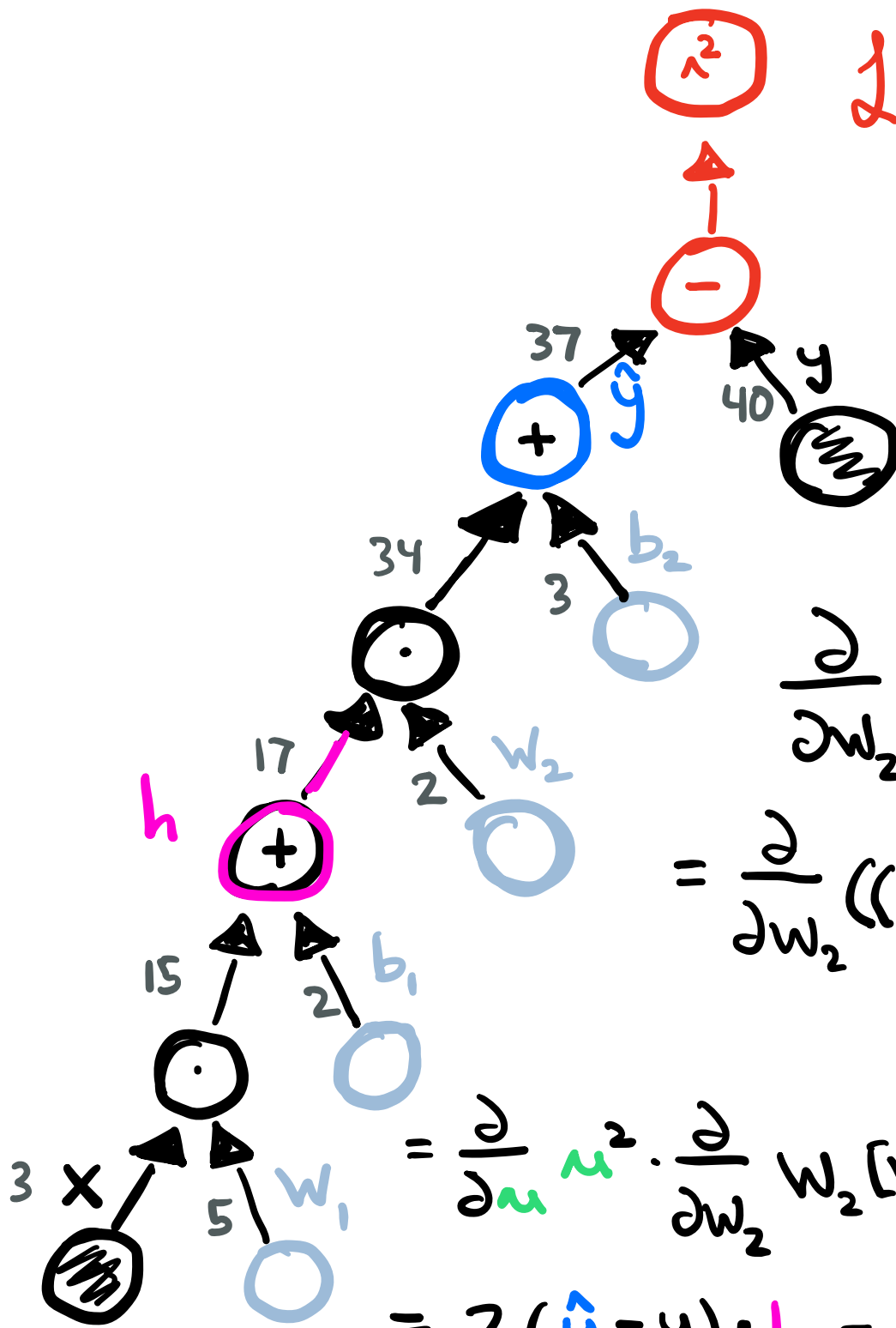
Let's build a **graph** for a simple **regression** model

$$\hat{y} = w_2 \cdot (w_1 \cdot x + b_1) + b_2$$

$$\mathcal{L}(\hat{y}, y) = (\hat{y} - y)^2$$

Computation graph

Let $x=3; w_1=5; b_1=2; w_2=2; b_2=3; y=40$



$$\frac{\partial \mathcal{L}}{\partial w_2} = \frac{\partial}{\partial w_2} (\hat{y} - y)^2$$

$$= \frac{\partial}{\partial w_2} ((w_2 \cdot [w_1 \cdot x + b_1] + b_2) - y)^2$$

$$= \frac{\partial}{\partial u} u^2 \cdot \frac{\partial}{\partial w_2} w_2 [w_1 \cdot x + b_1] + b_2 - y$$

$$= 2(\hat{y} - y) \cdot h = 2(37 - 40) \cdot 17$$

$$= -102$$

The local rule. The graph lets us think about 3 things for each node.

(1) ITS inputs

(2) ITS output(s)

(3) The derivative of (2) w.r.t. (1)

Consider node C. We saw 3 operations above.

$$C = a + b \quad \frac{\partial C}{\partial a} = \frac{\partial C}{\partial b} = 1$$

grad. passes through
to a, b

$$C = a \cdot b \quad \frac{\partial C}{\partial a} = b; \frac{\partial C}{\partial b} = a$$

multiplied by other input

$$C = u^2 \quad \frac{\partial C}{\partial u} = 2u$$

Gradient at a node.

$(\text{incoming gradient}) \times (\text{local } \frac{\partial}{\partial x})$

Let's go ← Backwards

$$\star \frac{\partial \mathcal{L}}{\partial f} = 1.$$

<u>Node</u>	<u>(incoming) x local</u>	<u>$\frac{\partial \mathcal{L}}{\partial}$</u>
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$\hat{y}$

$$1 \cdot 2(\hat{y} - y)$$

$$-6$$

b_2

$$-6 \cdot 1$$

$$-6$$

w_2

$$-6 \cdot h = -6 \cdot 17$$

$$-102$$

as above!

h

$$-6 \cdot w_2 = -6 \cdot 2$$

$$-12$$

$$\left(\frac{\partial}{\partial h} w_2 h = w_2 \right)$$

b_1

$$-12 \cdot 1$$

$$-12$$

w_1

$$-12 \cdot x = -12 \cdot 3$$

$$-36$$

x

$$-12 \cdot w_1 = -12 \cdot 5$$

$$-60$$

We'll do more ← next time!

(See notebook.)

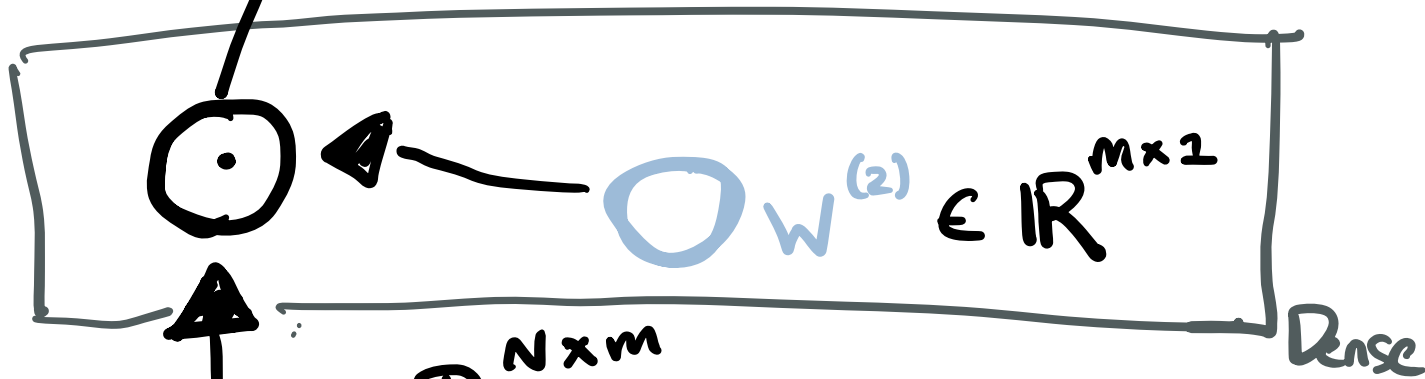
We have been considering Scalars
 but in general Should Think
 of **Tensors**.

$$\hat{y} = W^{(2)} \cdot \sigma(W^{(1)} \cdot x)$$

$y \in \mathbb{R}^N$

$\mathcal{L}$

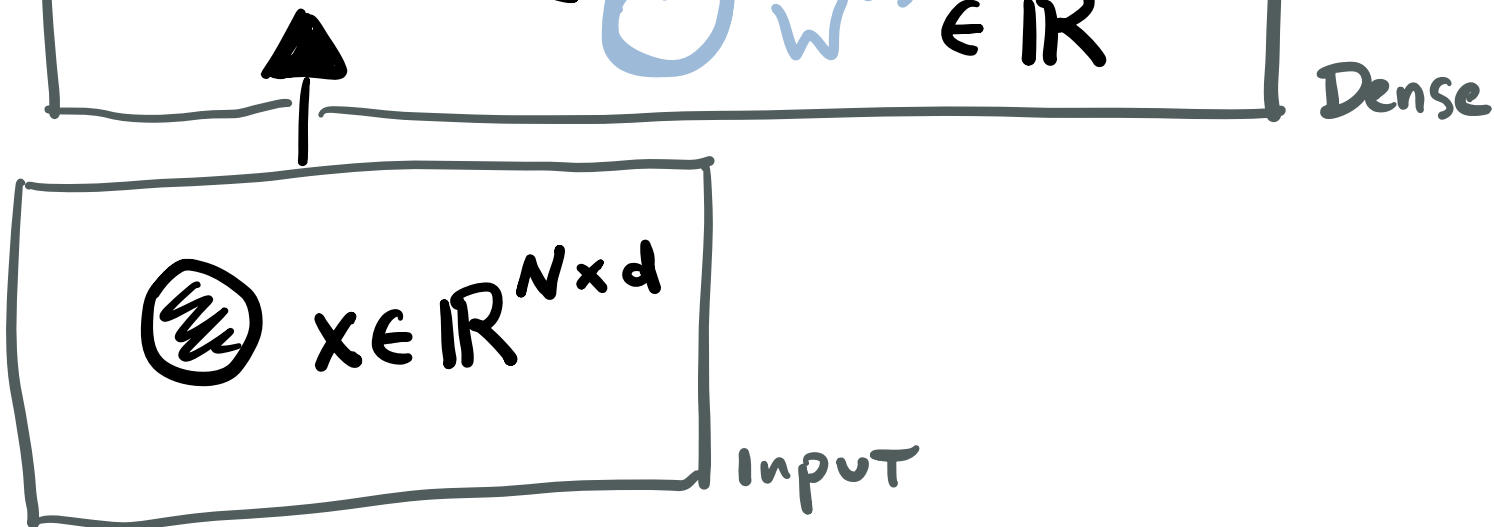
$\hat{y}$



σ

(element-wise)
 Activation





Gradient Shapes!

$\frac{\partial \mathcal{L}}{\partial W}$ will have same shape as W .

Consider layer $h = xW$

$$\frac{\partial \mathcal{L}}{\partial W} = x^T \frac{\partial \mathcal{L}}{\partial h}$$

$\underbrace{\langle d \times N \rangle \langle N \times m \rangle}_{\langle d \times m \rangle} \checkmark$

$$\frac{\partial \mathcal{L}}{\partial x} = \frac{\partial \mathcal{L}}{\partial h} W^T$$

$\underbrace{\langle N \times m \rangle \langle m \times d \rangle}_{\langle N \times d \rangle} \checkmark$

$$\left[\frac{\partial \mathcal{L}}{\partial W} \right]_{ij} = \sum_{n=1}^N \underbrace{x_{in}^T}_{\text{How changing } W_i \text{ in } n\text{-th ex. changes } h_j} \left[\frac{\partial \mathcal{L}}{\partial h} \right]_{nj}$$

How much loss h_j is responsible for

Still (incoming) $\times$ (local), but have to be careful with dimensions. More on this next lecture!

Note that this can be assembled **dynamically** i.e. before a **forward** pass.

In code, each block is a **Module** which is a class.

We can **chain** these together.